Teacher Pay, Skill Premia, and Long-Run Inequality*

Anson Zhou†

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Abstract

Policies that compress teacher pay are often motivated by a desire for equity. This paper shows that, in general equilibrium, such policies can backfire. A heterogeneous-agent overlapping-generations model demonstrates that compressing teacher pay reduces the supply of education services, lowers future human capital, and raises the skill premium, which disproportionately benefits high-skill workers. In the long run, income inequality rises, and the largest losses in human capital fall on children from low-income families. By contrast, tightening teacher qualification standards or expanding the teacher workforce reduces long-run inequality, even though each of these reforms appears regressive in the short run. These sign reversals underscore the necessity of a micro-to-macro approach to education policy evaluation.

JEL classification: I24, J24, J31, J45

Keywords: teacher pay, human capital, inequality, general equilibrium

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1 Introduction

Teachers represent less than 5% of the workforce, yet they shape the human capital of every subsequent generation. A large empirical literature has demonstrated that teacher quality is among the strongest school-based predictors of students' academic achievement and long-run economic outcomes (Rockoff, 2004; Rivkin et al., 2005; Chetty et al., 2014a). The rules that govern teacher compensation—and, more broadly, the way societies allocate talent between teaching and other occupations—therefore have consequences that extend far beyond the classroom (Jackson et al., 2014).

In many countries, teacher pay is determined by rigid schedules that tie salaries almost exclusively to seniority and academic credentials, with little or no reward for effectiveness. Collective bargaining agreements often reinforce this compression, formally preventing school districts from differentiating pay by performance (Hoxby and Leigh, 2004; Biasi, 2021). These arrangements are frequently defended on equity grounds: they limit pay disparities among teachers and, by insulating placement from market forces, are thought to promote a more uniform distribution of educational resources across schools and neighborhoods.

This paper argues that the general equilibrium logic behind that defence is flawed. When returns to skill in the teaching profession are compressed, high-ability individuals have weaker incentives to enter or remain in the classroom (Lavy, 2002; Tincani, 2021). The average quality of education services declines, and the human capital of the next generation erodes. Over time, the scarcity of human capital in the broader labor market bids up the skill premium (the price of human capital), which disproportionately benefits high-skill workers and widens the income distribution. Additionally, children from low-income families, who rely most heavily on public education, suffer the largest losses in human capital when teaching resources become scarce. In the long run, a policy designed to make teacher pay more equal can therefore increase inequality in the economy as a whole.

To quantify these mechanisms, I develop a heterogeneous-agent overlapping-generations (HA-OLG) model with occupational choice and dynastic human capital formation. In the model, altruistic adults decide whether to become teachers or workers, how much effort to supply, and how many education services to purchase for their children. The government regulates four dimensions of the education sector: the number of teaching positions, the minimum human capital threshold for entering the profession, the return to human capital within teaching (performance pay), and the progressivity with which education services are allocated across households. Because teachers are recruited from the adult population and their effectiveness determines the human capital of future cohorts, the quantity and quality of the teaching workforce and the entire human capital distribution co-evolve over time.

I use the model to trace three channels through which a decline in the supply of education services—driven, for example, by teacher pay compression or improving outside options—amplifies inequality. First, a *price channel* operates through the skill premium: when aggregate human capital falls, its price rises, favouring workers who are already highly skilled. Second, a *quantity channel* operates through the allocation of education services: when these services become scarcer, richer parents out-compete poorer ones, widening the dispersion of human capital in the next generation. Third, a *dynamic channel* captures the compounding of teacher quality over successive cohorts: today's weaker teachers produce tomorrow's weaker labor force, from which the next wave of teachers will be drawn. Because the dynamic channel takes time to play out, a policy's long-run effect can carry the *opposite sign* from its short-run, partial-equilibrium impact. I will examine these effects using four policy experiments.

I calibrate the model to U.S. data, matching moments from the Current Population Survey (CPS), the National Longitudinal Survey of Youth (NLSY), and published estimates of teacher value-added (Chetty et al., 2014a) and intergenerational mobility (Chetty et al., 2014b). The calibrated model reproduces several non-targeted empirical patterns, most notably the differential damage that teacher wage compression inflicts on children from low-income households, a finding reported by Lovenheim and Willén (2019) and Lavy (2020).

Four policy experiments illustrate the trade-offs that the framework uncovers. (1) Compressing teachers' performance pay—the model counterpart of a duty-to-bargain law that flattens salary schedules—raises the income Gini coefficient by 0.9% in the long run, with children at the bottom of the distribution losing three times as much human capital as those at the top. (2) Making the allocation of education services more equitable reduces long-run inequality substantially, with effects that are more than twice as large as those captured by a one-generation analysis. (3) Raising the minimum human capital threshold for teachers (a stand-in for stricter licensure) *increases* the Gini in the next generation but *decreases* it in the long run, as the initial crowding-out of high-skill workers gives way to a higher-quality education system. (4) Expanding the number of teachers exhibits the same sign reversal: short-run inequality rises, but long-run inequality falls.

The central message is that dynamic general equilibrium feedback from education policy is large enough to overturn standard static intuition. Policies designed to narrow pay differences inside the teaching profession can become self-defeating once their effects on the human capital of future workers—and therefore on the entire wage distribution—are taken into account. Conversely, reforms that appear costly or regressive in the short term may prove to be the most equitable levers over a long horizon. Methodologically, the results caution against evaluating education policies solely through the lens of contemporaneous, partial-equilibrium estimates, and they highlight the value of embedding micro evidence on teacher effectiveness within a macro framework that closes the loop between the education sector and the labor market.

Related Literature

A large empirical literature documents that teacher quality strongly affects students' long-run outcomes (Rockoff, 2004; Rivkin et al., 2005; Chetty et al., 2014a). Rigid step-and-lane schedules compress the link between pay and effectiveness, driving high-ability individuals out of teaching (Hoxby, 1996; Hoxby and Leigh, 2004; Corcoran et al., 2004; Bacolod, 2007). Flexible pay improves teacher sorting and student achievement, especially for disadvantaged groups (Lavy, 2002, 2020; Biasi, 2021), and compulsory pay compression inflicts lasting damage on children from low-income families (Lovenheim and Willén, 2019). These studies, however, analyze the teacher labor market in partial equilibrium and do not trace the effects on the broader skill premium and income distribution.

A parallel macro-literature studies education finance, human capital accumulation, and inequality. Fernandez and Rogerson (2003) and Zheng and Graham (2022) show that the progressivity of school funding shapes intergenerational mobility, while De la Croix and Doepke (2004) and Kotera and Seshadri (2017) endogenize the human capital distribution. Other work examines spatial or place-based education policies (Chyn and Daruich, 2022; Park and Hahm, 2023). In all of these models, however, teacher quality is either exogenous or a simple function of spending; occupational choice and the incentive structure of the teaching profession are absent.

A third strand explicitly models the supply side of the teacher market. Gilpin and Kaganovich (2012) and Tincani (2021) study teacher selection, but without an endogenous skill premium or economy-wide human capital feedback. More broadly, the allocation-of-talent literature emphasizes that reward structures in critical occupations can have aggregate consequences (Murphy et al., 1991; Acemoglu, 1995; Baumol, 1996; Gottlieb et al., 2023).

The paper closest to mine is Artige and Cavenaile (2023), who build an endogenous growth model with occupational choice among workers, teachers, and managers. They show that education quality depends on which agents select into teaching, and that the relationship between public education spending and inequality can be positive or negative. My framework differs in several respects. I focus on the *pay structure* rather than the level of spending, and I allow households to obtain different amounts of education services according to their income. Additionally, the skill premium is endogenously determined in a competitive labor market rather than by a span-of-control technology. These features allow me to isolate a price channel for inequality that operates through the skill premium, to capture the quantity channel that arises when richer parents out-compete poorer ones for scarcer services, and to document sign reversals between short-run and long-run policy effects that do not appear in steady-state comparisons.

The present paper therefore integrates the micro evidence on teacher pay (Biasi, 2021; Lovenheim and Willén, 2019) with the macro literature on education and inequality (Zheng and Graham, 2022; Kotera and Seshadri, 2017), providing a tractable framework in which the long-run, general-equilibrium effects of education policies—including sign reversals relative to partial-equilibrium estimates—can be quantified.

2 Model

I build a heterogeneous-agent overlapping-generations model in which altruistic parents divide their time between work and child-rearing, choose between two occupations, and purchase education services for their children. The model is deliberately parsimonious, isolating the interplay between the teacher labor market and the evolution of the economy-wide human capital distribution.

2.1 Households

The unit of analysis is a dynasty. Each adult lives for one period, bears exactly one child, and makes all decisions. Adults differ solely by their human capital $h \in \mathbb{R}_{++}$.

An adult chooses an occupation $o \in \{\mathcal{W}, \mathcal{T}\}$ (worker or teacher), work effort $l \geq 0$, consumption $c \geq 0$, and the amount of education services $e \geq 0$ that their child will receive. Utility is dynastic:

$$V(h) = \max_{o,l,c,e} \log(c) - \mu \frac{l^{1+1/\nu}}{1+1/\nu} + \delta\, \mathbb{E}_\epsilon\, V(h'), \tag{1}$$

where $\mu > 0$ scales disutility of effort, $\nu > 0$ is the Frisch labor supply elasticity, $\delta \in (0,1)$ is the altruism weight, and ϵ is an idiosyncratic ability shock. Flow utility is additively separable in the style of King, Plosser, and Rebelo.

The budget constraint is

$$c = y_o(h,l)\,(1-\tau), \tag{2}$$

where τ is a linear income tax. Income in each occupation follows

$$y_o(h,l) = \begin{cases} \beta_{\mathcal{W}} + w_{\mathcal{W}}\, h\, l, & o = \mathcal{W}, \\ \beta_{\mathcal{T}} + w_{\mathcal{T}}\, h\, l, & o = \mathcal{T}, \end{cases} \tag{3}$$

with β_o a base remuneration for labor input and w_o the piece rate per efficiency unit $h\,l$. I assume throughout that $w_{\mathcal{T}} < w_{\mathcal{W}}$, a condition verified in the data (Section 3 and Appendix B). The piece-rate gap captures the central institutional feature: teaching compresses the returns to human capital relative to the broader labor market.

The child's human capital is produced according to

$$h' = \epsilon\, h^\rho\, e^\gamma, \qquad \log\epsilon \sim \mathcal{N}\left(-\frac{\sigma_\epsilon^2}{2}, \sigma_\epsilon^2\right). \tag{4}$$

The parameter $\rho \in (0,1)$ governs the intergenerational transmission of human capital through non-market channels (parenting, genetics, environment), while $\gamma \in (0,1)$ measures the elasticity of child human capital with respect to formal education services. The multiplicative structure implies that parental background and school inputs are complements, a specification widely used in the macro-human-capital literature (De la Croix and Doepke, 2004; Zheng and Graham, 2022).

Education services e are provided by the government but are allocated according to a parametric rule that depends on parental income:

$$e = p\,\big(y_o(h,l)\,\tau\big)^{\eta}, \qquad \eta > 0. \tag{5}$$

The scalar p is an endogenous shadow price that clears the market for education services; it can be thought of as the inverse of the per-pupil price that schools face when converting tax revenues into teacher–student contact time. The parameter η is the primary policy lever for the progressivity of education service allocation. When η is low, the link between family income and resources is weak (progressive); when η is high, richer families command disproportionately more services (regressive). In the United States, local property taxes finance over half of public school expenditures (Zheng and Graham, 2022), which justifies tying e to own income; the functional form captures the essential correlation between family resources and school quality without explicitly modelling residential choice or multiple school districts.

A critical feature of the household problem is that parents internalise the feedback from their own effort and occupational choice to their child's human capital. Because a higher own income y_o raises the child's e through (5), which in turn increases h' through (4), a parent's choice improves the expected utility of the next generation. Altruistic parents therefore work harder and choose occupations partly to improve their children's opportunities.[1] At the same time, they take all aggregate prices and the tax rate as given.

2.2 Firms and Schools

A representative firm produces the consumption good with a technology that is constant returns to scale (CRS) and features constant elasticity of substitution (CES):

$$C = \left[\lambda L^{\frac{\phi-1}{\phi}} + (1-\lambda) H^{\frac{\phi-1}{\phi}}\right]^{\frac{\phi}{\phi-1}}, \tag{6}$$

[1] I abstract from an explicit choice of private parental investments such as tutoring, books, or parental time. Introducing an additional parental investment choice would create a standard crowding-out effect: when public education services deteriorate, parents partially compensate by increasing their own investments. The quantitative response would therefore be dampened, consistent with the findings of Kotera and Seshadri (2017) and Daruich (2018). The *qualitative* mechanism that this paper highlights—that lower teacher quality reduces future human capital and raises the skill premium—remains robust, as long as parental and public inputs are not perfect substitutes.

where L is aggregate raw labor (the number of workers) and H is aggregate human capital supplied by workers:

$$L = \int_{\Theta_{\mathcal{W}}} dF(h), \qquad H = \int_{\Theta_{\mathcal{W}}} l^*(h)\, h\, dF(h). \tag{7}$$

The parameter $\phi > 1$ is the elasticity of substitution between labor and human capital. Empirically, the consensus estimate for ϕ lies in the range 1.5–3 (Katz and Murphy, 1992; Bils et al., 2024).

A representative school produces divisible education services by employing teachers. I assume a linear technology: one efficiency unit of teacher time, $h\,l$, yields one unit of education services. The total supply of education services is therefore

$$E = \int_{\Theta_{\mathcal{T}}} l^*(h)\, h\, dF(h). \tag{8}$$

All services are allocated across households according to (5), and p adjusts so that total demand equals total supply:

$$\int p\,(y^*(h)\tau)^{1-\eta}\, dF(h) = E. \tag{9}$$

2.3 Government

The government operates with a balanced budget and uses four instruments to shape the teacher labor market and the education market:

(1) **Number of teaching positions.** The government posts Ω teaching slots, which are filled in equilibrium. Because teachers enjoy a tenure-like guarantee, Ω also fixes the size of the teaching workforce.

(2) **Minimum human capital rank.** Only individuals whose human capital ranks at or above κ_h may choose $o = \mathcal{T}$. That is, if $F(\cdot)$ denotes the cumulative distribution of human capital, then $F(h) \geq \kappa_h$ is a prerequisite for teaching. In practice, κ_h captures licensure requirements, subject-matter competency tests, or statutory minimum qualifications.

(3) **Return to human capital in teaching.** The government sets $w_{\mathcal{T}}$, the piece rate for teacher efficiency units. A lower $w_{\mathcal{T}}$ implies greater compression of teacher pay; it is the model's representation of the rigid salary schedules enforced by collective bargaining agreements (Biasi, 2021). I abstract from the political process that determines $w_{\mathcal{T}}$ and treat it as an exogenous policy parameter, a common practice in macro-public-finance models.

(4) **Progressivity of education service allocation.** The parameter η determines how strongly a household's education services depend on its tax contribution. Policies such as school finance equalization, income-dependent vouchers, or place-based interventions shift η.

The income tax rate τ adjusts endogenously to satisfy

$$\int_{\Theta_{\mathcal{T}}} y^*(h)\, dF(h) = \int \tau\, y^*(h)\, dF(h). \tag{10}$$

These four instruments provide a parsimonious yet comprehensive representation of the main policy levers through which governments affect teacher quality and the distribution of education resources. The number of teaching positions Ω captures class-size mandates and teacher–student ratio regulations, which have been shown to influence student achievement and teacher demand (Browning and Heinesen, 2007; Chingos and Whitehurst, 2011). The minimum human capital rank κ_h mimics occupational licensing and entry barriers that alter the composition of the teacher workforce (Wiswall, 2007). The piece rate $w_{\mathcal{T}}$ summarizes the extent of performance pay versus rigid salary schedules, reflecting the degree of wage compression that arises from collective bargaining or duty-to-bargain laws (Hoxby and Leigh, 2004; Biasi, 2021). Finally, the progressivity parameter η corresponds to the degree of school finance equalization; a lower η implies a more progressive allocation, akin to reforms that weaken the link between local property taxes and per-pupil spending (Fernandez and Rogerson, 2003; Zheng and Graham, 2022). Together, these four parameters span the primary policy dimensions that empirical work has identified as crucial for teacher selection and student outcomes, while keeping the model tractable.

2.4 Equilibrium Definition

Let Θ_o be the set of individuals choosing occupation o. A stationary competitive equilibrium consists of policy functions $\{o^*(h), l^*(h), c^*(h), e^*(h)\}$, a human capital distribution $F(h)$, prices $\{\beta_{\mathcal{W}}, w_{\mathcal{W}}, \beta_{\mathcal{T}}, p\}$, a tax rate τ, and the occupational interval $\Theta_{\mathcal{T}} = [\underline{h}, \overline{h}]$ such that:

1. *Household optimization.* Given prices and τ, the policy functions solve (1)–(5).
2. *Firm optimization.* The firm chooses L and H so that factor prices equal marginal products: $\beta_{\mathcal{W}} = \partial C/\partial L$, $w_{\mathcal{W}} = \partial C/\partial H$.
3. *Teacher labor market.* $\Omega = \int_{\Theta_{\mathcal{T}}} dF(h)$ and (9) hold.
4. *Worker labor market.* L and H are given by the integrals over $\Theta_{\mathcal{W}}$.
5. *Goods market.* $\int c^*(h)\, dF(h) = C$.
6. *Government budget.* Equation (10) holds.
7. *Stationarity.* The human capital distribution reproduces itself:

$$F(h_k) = \iint \mathbf{1}\big(\epsilon\, h^{\rho}\, (e^*(h))^{\gamma} < h_k\big)\, d\mathcal{N}(\epsilon)\, dF(h). \tag{11}$$

Under the empirically verified condition $w_{\mathcal{T}} < w_{\mathcal{W}}$, the occupational partition takes the simple form depicted in Figure 1.

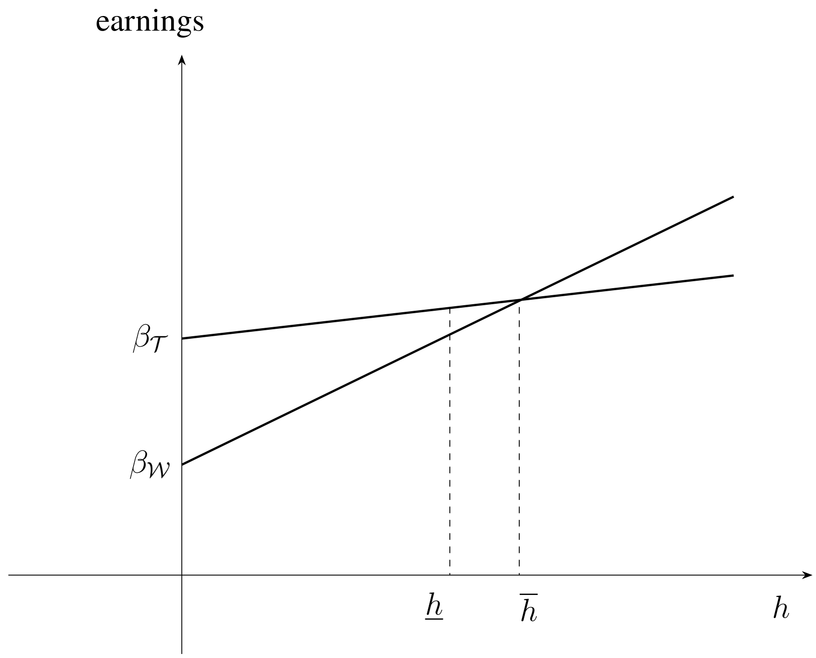

Figure 1. Occupational choice in the stationary equilibrium

Notes: The figure illustrates optimal occupational choices as a function of human capital h. The dashed vertical lines at $\underline{h}$ and $\overline{h}$ separate workers ($h < \underline{h}$), teachers ($h \in [\underline{h}, \overline{h}]$), and workers ($h > \overline{h}$). The intercepts $\beta_{\mathcal{W}}$ and $\beta_{\mathcal{T}}$ are the base wages, and the slopes are $w_{\mathcal{W}}$ and $w_{\mathcal{T}}$, with $w_{\mathcal{T}} < w_{\mathcal{W}}$ reflecting teacher pay compression.

Individuals with human capital below $\underline{h}$ are barred from teaching by the rank requirement. Individuals with human capital above $\overline{h}$ find the private sector more remunerative, as the slope $w_{\mathcal{W}}$ exceeds $w_{\mathcal{T}}$. The bounds $\underline{h}$ and $\overline{h}$ satisfy $F(\underline{h}) = \kappa_h$, $F(\overline{h}) - F(\underline{h}) = \Omega$, and income equalization at the upper threshold: $\beta_{\mathcal{T}} + w_{\mathcal{T}}\overline{h}\, l^*(\overline{h}) = \beta_{\mathcal{W}} + w_{\mathcal{W}}\overline{h}\, l^*(\overline{h})$.

The equilibrium can be computed by value function iteration combined with a fixed-point search on the distribution $F(h)$, the factor prices, and the education shadow price p. The numerical algorithm is described in Appendix A.

2.5 Mechanisms

I now outline how a compression of $w_{\mathcal{T}}$ propagates through the economy and raises inequality. These channels are also triggered by any policy that reduces the supply of education services.

Price channel. A lower $w_{\mathcal{T}}$ reduces the effective remuneration of high-skill teachers. Those near the upper threshold $\overline{h}$ re-optimize their occupation choice: some become workers, shrinking the set $\Theta_{\mathcal{T}}$ and lowering average teacher quality. Because teachers are the sole input into education services, aggregate services E fall. Through h', this lowers the human capital of the next generation. Scarcer human capital in the labor market forces up the skill premium $w_{\mathcal{W}}$. Individuals with higher h earn a larger share of their income from the efficiency-unit component $w_{\mathcal{W}}\, h\, l$, so the rise in $w_{\mathcal{W}}$ disproportionately benefits them, widening the cross-sectional income distribution.

Quantity channel. When E contracts, the equilibrium shadow price p rises, intensifying competition for education services. Because of the allocation rule (5), richer parents can bid up their effective demand more easily, obtaining a larger fraction of the diminished total supply. The dispersion of e across households increases, which, via $\gamma > 0$, translates into a wider human capital distribution in the next generation. This quantity effect reinforces the price channel: a wider human capital distribution raises the skill premium further and amplifies inequality.

Dynamic channel. Teachers are not an exogenous endowment; they are drawn from the adult human capital distribution. Today's lower teacher quality reduces the human capital of tomorrow's workers, from which tomorrow's teachers will be drawn. Over multiple generations, the decline in teacher quality compounds. The dynamic channel implies that the long-run inequality response to a policy change can be several times larger than what a one-generation evaluation would suggest.

These three channels interact. A higher skill premium raises the opportunity cost of teaching, accelerating the exit of high-skill individuals. The static equalizing intent of pay compression is thus undermined by the very dynamics it sets in motion.

3 Calibration

I calibrate the model to reflect the U.S. economy around the year 2000. The parameters are partitioned into two groups: those taken from the existing literature and those calibrated internally by matching data moments.

3.1 Externally Set Parameters

Parental altruism δ is fixed at 0.34, the value used by Daruich (2018) in a closely related OLG-human-capital setting. The Frisch elasticity of labor supply ν is set to $1/3$, a standard value in the macro-labor literature (Keane, 2011). The intergenerational persistence parameter ρ is set to 0.27, based on the estimates of Lefgren et al. (2012) for the parent–child earnings correlation net of schooling effects. The elasticity of substitution between raw labor and human capital in goods production, ϕ, is set to 3.0, a midpoint between the estimates of Katz and Murphy (1992) and Bils et al. (2024).

3.2 Internally Calibrated Parameters

The remaining parameters $\{\mu, \gamma, \sigma_\epsilon, \eta, \kappa_h, \Omega, w_{\mathcal{T}}, \lambda\}$ are chosen so that the stationary equilibrium matches a set of empirical moments. Although all parameters affect all moments in a general equilibrium system, each moment is most informative about a particular parameter. I describe the identification logic parameter by parameter.

Disutility of effort μ. Average hours worked in the CPS-ASEC for prime-age workers are approximately 0.3 (i.e., 30% of the time endowment). I set μ so that the model reproduces this average effort level.

Teacher value-added elasticity γ. Chetty et al. (2014a) estimate that a one-standard-deviation increase in teacher quality (as measured by test-score value-added) raises students' annual earnings in adulthood by 1.3%. Aggregating over twelve years of compulsory schooling gives a cumulative earnings increment of approximately 15.6%. In the model, I compute the standard deviation of education services e across households and, using $h' = \epsilon h^\rho e^\gamma$, calculate the effect of providing the median child with an additional one-standard-deviation of e. I adjust γ via indirect inference until the implied earnings gain equals 15.6%.

Ability shock dispersion σ_ϵ. The Gini coefficient of gross earnings in the CPS-ASEC is 0.42. In the model, σ_ϵ is the principal determinant of cross-sectional human capital dispersion. I calibrate it to match the Gini exactly.

Progressivity of education services η. Chetty et al. (2014b) report that the expected income rank of a child whose parents are at the 25th percentile of the national income distribution is 0.42. I calibrate η so that the model reproduces this absolute upward mobility statistic. A larger η makes the allocation of e more regressive, increasing income persistence; thus η is strongly identified by this moment.

Teacher human capital threshold κ_h. Using the NLSY79, I compute the AFQT percentile rank of the typical teacher. Because the AFQT score correlates strongly with earnings and is a standard proxy for cognitive human capital, I target the fact that the 25th percentile of the AFQT distribution among teachers corresponds to the 60th percentile of the overall distribution (i.e., $\kappa_h = 0.6$).

Number of teaching positions Ω. The share of the labor force employed as teachers (OCC90LY codes 155–163 in the CPS) is approximately 4%. I set Ω so that the model reproduces this share.

Return to human capital in teaching $w_{\mathcal{T}}$. To calibrate the wedge between teaching and non-teaching returns to skill, I regress log hourly wages on AFQT percentile separately for teachers and non-teachers in the NLSY79. The ratio of the two gradients is 0.75 (see Appendix B for details). With $w_{\mathcal{W}}$ determined by the firm's first-order condition, this ratio pins down $w_{\mathcal{T}}$.

Labor weight in production λ. Heathcote et al. (2023) report that the 90–10 ratio of household labor income in the United States is approximately 9. In the model, λ influences the skill premium and therefore the spread of earnings: a larger λ reduces the demand for human capital, compressing the income distribution. I calibrate λ to match the 90–10 ratio.

3.3 Calibration Results

Table 1 summarizes the parameter values and their empirical targets. The model matches all targeted moments exactly by construction. It also performs well on two important non-targeted dimensions. First, the model generates a differential response of child human capital to teacher wage compression that aligns with the quasi-experimental evidence: Lovenheim and Willén (2019) show that Black and Hispanic students suffer larger long-term losses from exposure to collectively bargained teacher pay scales, and Lavy (2020) finds that performance-pay programs disproportionately benefit children from below-median income families. In my model, compressing $w_{\mathcal{T}}$ reduces the expected human capital of children at the 10th percentile three times as much as those at the 90th percentile in the long run, consistent with these patterns. Second, the model-implied distribution of teacher AFQT ranks and the teacher–non-teacher wage gradient are mutually consistent with the NLSY evidence, providing an internal validation of the occupational choice mechanism.

4 Policy Counterfactuals

I now conduct four counterfactual experiments that illustrate the versatile policy levers the framework can accommodate. Each experiment is run under three informational scenarios: (1) the *long-run stationary general equilibrium*, in which both the human capital distribution and the skill premium adjust fully; (2) a *fixed human capital distribution*, which isolates the contemporaneous general equilibrium price effects; and (3) a *next-generation only* calculation, which captures the outcome after a single period of adjustment. Differences among the three reveal the quantitative importance of endogenous prices and the slow-moving nature of human capital.

The four experiments are not arbitrary parameter shifts; each one corresponds to a specific, widely studied education policy. The experiment on the distribution of education services (η) speaks to the large literature on school finance equalization and the consequences of funding formulas that tie resources to local property values (Fernandez and Rogerson, 2003; Zheng and Graham, 2022). The compression of $w_{\mathcal{T}}$ directly mimics the effect of duty-to-bargain laws and collective bargaining agreements that flatten the relationship between teacher pay and teacher effectiveness, as documented by Hoxby and Leigh (2004) and Biasi (2021). The change in κ_h captures the long-standing policy debate over teacher licensure and certification requirements, a debate in which Wiswall (2007) provides key empirical evidence on the trade-off between teacher quality and quantity. Finally, the increase in Ω reflects class-size reduction mandates, a perennial policy proposal whose effects on student achievement remain contested (Browning and Heinesen, 2007; Chingos and Whitehurst, 2011). In each case, the direction and magnitude of the parameter

Table 1. Calibration outcomes

Parameter	Interpretation	Value	Target moment	Source	Fit
			Externally set		
δ	Altruism	0.34	–	Daruich (2018)	–
ν	Labor supply elasticity	0.33	–	Keane (2011)	–
ρ	Parental spillover	0.27	–	Lefgren et al. (2012)	–
ϕ	Substitutability (L vs. H)	3.0	–	Katz and Murphy (1992); Bils et al. (2024)	–
			Internally calibrated		
μ	Effort disutility	25	Avg. hours = 0.3	CPS-ASEC	0.3
γ	Teacher value added	0.11	1 SD VA $\Rightarrow$ 1.3% annual earnings	Chetty et al. (2014a)	1.3%
σ_ϵ	Ability shock dispersion	2.3	$\text{Gini}_y = 0.42$	CPS-ASEC	0.42
η	Education service elasticity	2.5	Rank mobility = 0.42	Chetty et al. (2014b)	0.42
κ_h	Teacher qualification rank	0.6	Teacher AFQT rank = 0.6	NLSY	0.6
Ω	Number of teachers	0.04	Teacher share = 0.04	CPS-ASEC	0.04
$w_{\mathcal{T}}$	Performance pay coefficient	2.5	Relative AFQT gradient = 0.75	NLSY	0.75
λ	Labor weight in CES	0.94	90/10 income ratio $\approx$ 9	Heathcote et al. (2023)	9.0

Notes: The table reports the calibrated parameter values and the empirical targets used in the estimation. Externally set parameters are taken from the cited literature. Internally calibrated parameters are chosen to match the listed moments from U.S. data around 2000. The model is just identified and matches each target exactly. The last column reports the model fit for the targeted moments. AFQT denotes the Armed Forces Qualification Test score from the NLSY79.

change is calibrated to match the order of magnitude of the corresponding real-world policy variation.

4.1 More Equitable Distribution of Education Services

The first experiment lowers η from 2.5 to 2.0, a 20% reduction that makes the education service allocation less sensitive to parental income. This magnitude approximates the change in the elasticity of education spending to family income that would occur if a state moved from a strongly local-property-tax-based system to a heavily state-equalized one (Zheng and Graham, 2022).

Table 2 reports the results. In the long run, average human capital rises by 6.1%, the skill premium falls by 8.1%, and the income Gini drops by 4.5%. The 90–10 income ratio declines by 6.7%, while the 90–10 human capital ratio narrows by 5.1%. Children at the 10th percentile of the parent distribution gain over 10% in expected human capital, while those at the 90th percentile lose 1.4%—a clear redistributive effect.

Fixing $F(h)$ entirely eliminates the human capital gains and reverses the sign of the inequality change: both the Gini and the 90–10 ratio *increase* in this scenario because the skill premium rises when the reallocation of education services does not translate into a larger aggregate human capital stock. The next-generation scenario produces the right signs but quantitatively smaller effects: average human capital rises by only 2.7%, and the Gini falls by 1.6% rather than 4.5%. The dynamic amplification thus more than doubles the impact over the long run.

Table 2. More equitable distribution of education services

Change relative to baseline	Long-run GE	Fixed $F(h)$	Next gen.
Aggregate education services	+7.73%	−0.78%	+4.00%
Price of education services	−33.50%	−25.63%	−29.57%
Average human capital	+6.07%	0.00%	+2.66%
Skill premium	−8.14%	+1.04%	−2.58%
Income Gini	−4.53%	+0.63%	−1.59%
$90-10$ income ratio	−6.70%	+1.49%	−1.48%
$90-10$ human capital ratio	−5.05%	0.00%	−4.92%
$\mathbb{E}(h')$ at 10th pct.	+10.36%	+8.24%	+8.24%
$\mathbb{E}(h')$ at 90th pct.	−1.35%	−0.64%	−0.64%

Notes: The table reports percentage changes relative to the baseline stationary equilibrium. Long-run GE refers to the new stationary general equilibrium with fully adjusted human capital distribution and skill premium. Fixed $F(h)$ holds the human capital distribution constant, isolating the static general equilibrium price effects. Next gen. reports the outcome after one period, before the human capital distribution has fully adjusted. $\mathbb{E}(h')$ at the 10th (90th) percentile is the expected human capital of a child born to parents at the 10th (90th) percentile of the current human capital distribution.

4.2 Compressing Teacher Performance Pay

The second experiment compresses $w_{\mathcal{T}}$ by 20%, from 2.5 to 2.0. This is the model's counterpart of a duty-to-bargain law or a reinstatement of rigid step-and-lane salary schedules. The 20% reduction is similar in magnitude to the decline in pay dispersion that Biasi (2021) documents following the introduction of collective bargaining in some U.S. states. Table 3 shows that the policy depresses aggregate education services by roughly 10% in all scenarios; the effect is immediate because teachers reduce effort and some exit the profession.

In the long run, the skill premium rises by 1.5%, the income Gini increases by 0.9%, and the 90–10 income ratio climbs by 0.8%. The 90–10 human capital ratio actually *shrinks* by 1.5%, but this reflects a downward shift of the entire distribution, not an equalising compression within the upper tail. Critically, the loss of human capital is uneven: children at the 10th percentile lose 0.86%, while those at the 90th percentile lose only 0.28%. In the long run, this disparate impact widens further: the ratio of the 10th-percentile loss to the 90th-percentile loss is 1.8 in the next generation but grows to 3.1 in the steady state. Thus, a policy that appears to equalise pay *within* the teaching profession ends up deepening the human capital disadvantage of the children who most need high-quality schooling.

Table 3. Compressing teacher performance pay

Change relative to baseline	Long-run GE	Fixed $F(h)$	Next gen.
Aggregate education services	−9.67%	−7.97%	−9.65%
Price of education services	+5.53%	+4.89%	+5.76%
Average human capital	−0.97%	0.00%	−0.79%
Skill premium	+1.48%	+0.56%	+1.50%
Income Gini	+0.90%	+0.67%	+1.03%
$90-10$ income ratio	+0.77%	+1.12%	+1.61%
$90-10$ human capital ratio	-1.53%	0.00%	−0.76%
$\mathbb{E}(h')$ at 10th pct.	−0.86%	−0.70%	−0.70%
$\mathbb{E}(h')$ at 90th pct.	−0.28%	−0.39%	−0.39%

Notes: The table reports the effects of compressing the return to human capital in teaching, $w_{\mathcal{T}}$, by 20%. This mimics the compression of teacher pay scales through collective bargaining or duty-to-bargain laws. See Table 2 for scenario definitions. The long-run disparate impact ratio (loss at 10th percentile relative to loss at 90th percentile) is 1.8 in the next generation and grows to 3.1 in the long-run equilibrium.

4.3 Teacher Qualification Requirements

The third experiment raises the human capital rank threshold κ_h from 0.60 to 0.65. This corresponds to the licensure tightening that would shift the minimum AFQT rank of new teachers from the 60th to the 65th percentile of the college-educated distribution, akin to requiring a master's degree or a higher GPA (Wiswall, 2007). Because the same number of teaching slots must be filled from a narrower and higher-ability pool, the base wage $\beta_{\mathcal{T}}$—and the tax rate τ—rise.

Table 4 reveals a striking sign reversal. In the long run, aggregate education services surge by 18.6%, average human capital rises by 1.6%, and both the income Gini and the 90–10 income ratio *fall* (by 0.9% and 1.1%, respectively). The policy improves the human capital of children at the 10th percentile by 1.3% and by 0.7% at the 90th percentile, closing absolute gaps.

Yet the next-generation outcome paints the opposite picture: the Gini *rises* by 0.9% and the 90–10 ratio *rises* by 3.1%. The reason is a static crowding-out effect: when high-human-capital individuals switch to teaching, the immediate supply of human capital to the private sector shrinks, driving up the skill premium. High-skill non-teachers reap the gain, widening the contemporaneous income distribution. The fixed-$F(h)$ scenario, which omits any human capital accumulation, shows even larger inequality increases. The long-run benefits materialise only after the improved education services raise the human capital of the entire cohort—a process that takes several generations.

Table 4. Raising the teacher qualification requirement

Change relative to baseline	Long-run GE	Fixed $F(h)$	Next gen.
Aggregate education services	+18.59%	+15.92%	+17.38%
Price of education services	−9.37%	−6.02%	−6.12%
Average human capital	+1.64%	0.00%	+0.50%
Skill premium	−1.51%	+1.54%	+1.14%
Income Gini	−0.89%	+0.71%	+0.91%
$90-10$ income ratio	−1.11%	+2.73%	+3.13%
$90-10$ human capital ratio	−2.49%	0.00%	+0.76%
$\mathbb{E}(h')$ at 10th pct.	+1.29%	+0.34%	+0.34%
$\mathbb{E}(h')$ at 90th pct.	+0.70%	+1.09%	+1.09%

Notes: The table reports the effects of raising the human capital rank threshold for teachers from the 60th to the 65th percentile, akin to tightening licensure or education requirements for the profession. The tax rate rises endogenously to accommodate the higher base wages needed to attract higher-skill teachers. The sign reversal between the next-generation and long-run outcomes highlights the importance of accounting for endogenous human capital adjustment.

4.4 The Number of Teachers

In the final experiment, I raise the share of teachers in the labor force from 4% to 5%, equivalent to a class-size reduction from roughly 25 to 20 students per teacher in a typical U.S. elementary school (Chingos and Whitehurst, 2011). The policy is not targeted at the *quality* of teachers but at the *quantity* of teacher–student contact.

Table 5 again shows a sign reversal. Aggregate education services expand by 19%, and in the long run the income Gini falls by 0.5% and the 90–10 income ratio by 2.5%. Children at the 10th percentile gain 1.6% in expected human capital, compared to 0.6% at the 90th percentile.

In the immediate aftermath, however, the Gini *rises* by 0.8% (next generation) because the diversion of workers into teaching contracts the supply of human capital used in production, raising the skill premium. The fixed-$F(h)$ column confirms that this crowding-out channel is responsible. It is only once the expanded education services feed through into the skill distribution that the equalising forces dominate.

Table 5. Larger number of teachers

Change relative to baseline	Long-run GE	Fixed $F(h)$	Next gen.
Aggregate education services	+19.08%	+17.65%	+17.72%
Price of education services	-10.65%	-8.20%	-8.02%
Average human capital	+1.64%	0.00%	+0.61%
Skill premium	-2.61%	+0.26%	-0.46%
Income Gini	-0.54%	+0.84%	+0.82%
$90-10$ income ratio	-2.50%	+0.74%	+0.79%
$90-10$ human capital ratio	-2.49%	0.00%	+0.76%
$\mathbb{E}(h')$ at 10th pct.	+1.58%	+0.79%	+0.79%
$\mathbb{E}(h')$ at 90th pct.	+0.62%	+0.99%	+0.99%

Notes: The table reports the effects of increasing the share of teachers in the labor force from 4% to 5%, interpreted as a class-size reduction reform. The policy expands the quantity of education services but initially crowds out high-skill workers from the private sector. The long-run equalising effect materialises only after the increased education services raise the human capital of successive cohorts.

4.5 The Policy Irony in Summary

Table 6 distills the headline finding. Compressing teacher pay raises long-run inequality; tightening teacher standards and expanding the teacher workforce reduce long-run inequality but increase it in the short run. The sign flip between the next-generation and long-run outcomes cannot be detected by standard microeconometric methods that exploit single-generation variation. It arises because the initial labor-market reallocation and the subsequent human capital accumulation work in opposite directions, and the human capital adjustment is slow.

Table 6. Sign of inequality change

	Lower η	Compress $w_{\mathcal{T}}$	Raise κ_h	Raise Ω
Next-generation Gini	$-$	$+$	$+$	$+$
Long-run Gini	$-$	$+$	$-$	$-$

Notes: The table summarizes the direction of change in the income Gini coefficient under the four policy experiments analyzed in Section 4. A "+" denotes an increase relative to baseline, a "$-$" a decrease. The sign reversal between the next-generation and long-run outcomes for the qualification and quantity reforms underscores the importance of dynamic general equilibrium analysis.

The policy irony is stark: a reform that compresses teacher pay ends up increasing inequality economy-wide; reforms that raise teacher standards or increase the number of teachers are ultimately equalising. These results underscore the need to embed education policy analysis in a framework that accounts for the endogenous evolution of the human capital distribution and the skill premium.

5 Conclusion

This paper has developed a tractable HA-OLG model in which teacher quality, the human capital distribution, and the skill premium are jointly determined. Calibrated to U.S. data, the model demonstrates that policies often defended on equity grounds—most notably, compressing teacher performance pay—can increase long-run income inequality. By weakening incentives for high-skill workers to enter teaching, pay compression reduces the supply of education services, lowers future human capital, and raises the economy-wide skill premium. The consequence is a wider income distribution and a concentration of human capital losses among children from low-income families.

Conversely, policies that tighten teacher qualification requirements or increase the number of teachers reduce long-run inequality, even though they appear regressive in the short run. The sign reversal arises because the immediate crowding-out of skilled workers from production temporarily raises the skill premium, while the benefits of improved education services materialise only as the human capital distribution adjusts.

The framework can be extended in several directions. Adding a private education sector would allow households to opt out of public schools, potentially dampening the quantity channel but amplifying the price channel. Incorporating residential choice and property taxation would tighten the link between the model and school finance institutions. Embedding the static occupational choice into a life-cycle setting with human capital accumulation would enrich the teacher selection margin. Each of these extensions would further clarify the conditions under which education policies achieve their intended redistributive goals—or perversely undermine them.

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Appendix

A Computation Algorithm

The stationary equilibrium is computed by fixed-point iteration on the human capital distribution $F(h)$, the factor prices $\beta_{\mathcal{W}}, w_{\mathcal{W}}$, the teacher base wage $\beta_{\mathcal{T}}$, the tax rate τ, and the education shadow price p. The algorithm proceeds as follows:

1. **Initialization.** Guess $F(h)$, $\beta_{\mathcal{W}}$, $w_{\mathcal{W}}$, $\beta_{\mathcal{T}}$, τ, and p.

2. **Occupational partition.** Given κ_h and Ω, find $\underline{h}$ and $\overline{h}$ such that $F(\underline{h}) = \kappa_h$ and $F(\overline{h}) - F(\underline{h}) = \Omega$.

3. **Dynamic programming.** Solve the household's Bellman equation (1) by value function iteration on a grid of h, obtaining policy functions $o^*(h), l^*(h), c^*(h), e^*(h)$.

4. **Updating $F(h)$.** Use the human capital production function (4), the allocation rule (5), and the ability shock distribution to compute the implied distribution of h'. Smooth and update $F(h)$ with a dampening parameter.

5. **Updating prices.**

   (a) $\beta_{\mathcal{T}}$ is backed out from the income-equivalence condition $\beta_{\mathcal{T}} + w_{\mathcal{T}}\overline{h}\, l^*(\overline{h}) = \beta_{\mathcal{W}} + w_{\mathcal{W}}\overline{h}\, l^*(\overline{h})$.

   (b) New factor prices $\tilde{\beta}_{\mathcal{W}}$ and $\tilde{w}_{\mathcal{W}}$ are computed from the firm's marginal product conditions using the aggregates L and H.

   (c) The education shadow price p is updated from the market-clearing condition (9).

   (d) τ is adjusted to satisfy the government budget constraint (10).

6. **Convergence check.** Repeat steps 2–5 until the maximum change in $F(h)$, the prices, and τ falls below a tight tolerance.

The algorithm is parallelised across the inner value function iteration and the distributional update. Convergence is robust as long as the dampening parameter is chosen appropriately; typical runs require approximately 200 outer-loop iterations.

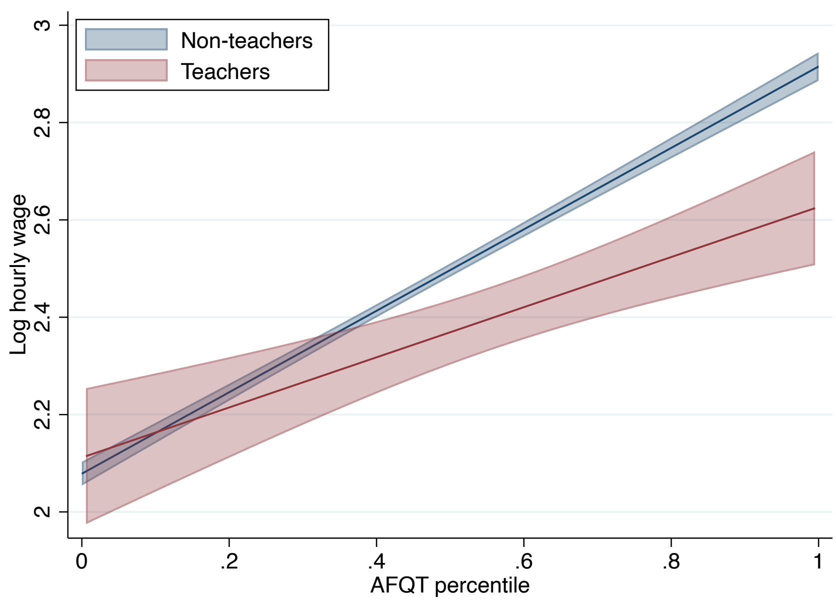
B Relative Returns to Human Capital

I estimate the ratio of the returns to human capital for teachers and non-teachers using the NLSY79. The AFQT (Armed Forces Qualification Test) percentile score, administered in 1980, serves as a proxy for cognitive human capital (Neal and Johnson, 1996). I restrict the sample to college-educated individuals working at least 30 weeks per year at an hourly wage of at least one dollar. The regression is

$$Y_{i,t} = \alpha_{j,t} + \Psi_{j,t}\,\mathrm{AFQT}_i + \varepsilon_{i,t}, \tag{12}$$

where $j = 1$ (teachers), $j = 2$ (non-teachers), t is the survey year (1996 or 2006), and $Y_{i,t}$ is the log hourly wage. Figure A.1 and Table A.1 report the results.

Figure A.1. AFQT and hourly wages



Notes: AFQT percentile score versus log hourly wage in 1996, NLSY79. Solid lines are linear fits; shaded areas are 90% confidence intervals. The sample is restricted to college-educated individuals working at least 30 weeks per year.

The gradient is consistently larger for non-teachers, and the ratio $\Psi_{1,t}/\Psi_{2,t}$ averages 0.754 across the two waves. I use this ratio in the calibration to set $w_{\mathcal{T}}/w_{\mathcal{W}} = 0.75$.

Table A.1. AFQT-wage regressions

	$t = 1996$		$t = 2006$	
	Teachers	Non-teachers	Teachers	Non-teachers
$\Psi_{j,t}$	0.515	0.837	0.827	0.927
	(0.113)	(0.024)	(0.122)	(0.029)
Observations	240	2,490	227	2,193

Notes: Robust standard errors in parentheses. The ratio of the AFQT gradient for teachers to that for non-teachers averages 0.754 across the two waves. This ratio is used to calibrate $w_{\mathcal{T}}/w_{\mathcal{W}}$ in the model.